

15-normal-subgroup: 正规子群 (Normal Subgroup)

魏恒峰

hfwei@hnu.edu.cn

2026 年 06 月 18 日



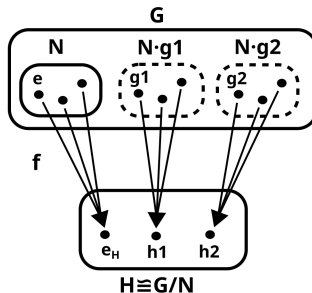
Fundamental Homomorphism Theorems



Emmy Noether (1882 ~ 1935)

Fundamental Homomorphism Theorems

Homomorphism theorem



● element of a group

□ group □ not a group

N normal subgroup

e identity of G e_H identity of H

f homomorphism $G \rightarrow H$

Help us understand the structure of a group
via its normal subgroups

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$(1)H = H = (1\ 2)H$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$(1)H = H = (1\ 2)H$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$H(1) = H = H(1\ 2)$$

$$H(1\ 3) = \{(1\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$H(2\ 3) = \{(2\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$(1)H = H = (1\ 2)H$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$H(1) = H = H(1\ 2)$$

$$H(1\ 3) = \{(1\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$H(2\ 3) = \{(2\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

It is possible that $aH \neq Ha$.

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$

$$(1)H = (1\ 2\ 3)H = (1\ 3\ 2)H = H \leq S_3$$

$$(1\ 2)H = (1\ 3)H = (2\ 3)H = \{(1\ 2), (1\ 3), (2\ 3)\}$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$

$$(1)H = (1\ 2\ 3)H = (1\ 3\ 2)H = H \leq S_3$$

$$(1\ 2)H = (1\ 3)H = (2\ 3)H = \{(1\ 2), (1\ 3), (2\ 3)\}$$

$$\forall a \in S_3 \ aH = Ha$$

Definition (Normal Subgroup (正规子群))

Suppose that $H \leq G$. If

$$\forall a \in G \ aH = Ha,$$

then H is a **normal subgroup** of G , denoted $H \triangleleft G$.

Definition (Normal Subgroup (正规子群))

Suppose that $H \leq G$. If

$$\forall a \in G \ aH = Ha,$$

then H is a **normal subgroup** of G , denoted $H \triangleleft G$.

$$aH = Ha \not\Rightarrow \forall h \in H \ ah = ha$$

Definition (Normal Subgroup (正规子群))

Suppose that $H \leq G$. If

$$\forall a \in G \ aH = Ha,$$

then H is a **normal subgroup** of G , denoted $H \triangleleft G$.

$$aH = Ha \not\Rightarrow \forall h \in H \ ah = ha$$

$$aH = Ha \Rightarrow \forall h \in H \ \exists h' \in H \ ah = h'a$$

Definition (正规子群的陪集)

Suppose that $H \triangleleft G$.

$$G/H = \{aH \mid a \in G\}$$

is the **set of cosets** of H in G .

Definition (Quotient Group (商群))

Suppose that $H \triangleleft G$. Define

$$aH \cdot bH = (ab)H.$$

Then $(G/H, \cdot)$ is a **group**, called the **quotient group** of G by H .

$aH \cdot bH = (ab)H$ is well-defined

$aH \cdot bH = (ab)H$ is well-defined

结果与代表元的选取无关

$aH \cdot bH = (ab)H$ is well-defined

结果与代表元的选取无关

$$aH = a'H \wedge bH = b'H \implies aH \cdot bH = a'H \cdot b'H$$

$aH \cdot bH = (ab)H$ is well-defined

结果与代表元的选取无关

$$aH = a'H \wedge bH = b'H \implies aH \cdot bH = a'H \cdot b'H$$

$$(1)H = H = (1\ 2)H$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$aH \cdot bH = (ab)H$ is well-defined

结果与代表元的选取无关

$$aH = a'H \wedge bH = b'H \implies aH \cdot bH = a'H \cdot b'H$$

$$(1)H = H = (1\ 2)H$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$(1\ 3)H \cdot (2\ 3)H = (1\ 2\ 3)H$$

$$(1\ 2\ 3)H \cdot (1\ 3\ 2)H = H$$

$aH \cdot bH = (ab)H$ is well-defined

结果与代表元的选取无关

$$aH = a'H \wedge bH = b'H \implies aH \cdot bH = a'H \cdot b'H$$

$aH \cdot bH = (ab)H$ is well-defined

结果与代表元的选取无关

$$aH = a'H \wedge bH = b'H \implies aH \cdot bH = a'H \cdot b'H$$

$$\begin{aligned} a'H \cdot b'H &= (a'b')H \\ &= a'(b'H) = a'(bH) = a'(Hb) \\ &= (a'H)b = (aH)b = a(Hb) = a(bH) \\ &= (ab)H = aH \cdot bH \end{aligned}$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$G/H = \{(1)H, (1\ 2)H\}$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$G/H = \{(1)H, (1\ 2)H\} \cong \mathbb{Z}_2$$

$$G = (\mathbb{Z}, +) \quad H = (6\mathbb{Z}, +) \triangleleft G$$

$$G/H =$$

$$G = (\mathbb{Z}, +) \quad H = (6\mathbb{Z}, +) \triangleleft G$$

$$G/H = \{0 + H, 1 + H, \dots, 5 + H\}$$

$$G = (\mathbb{Z}, +) \quad H = (6\mathbb{Z}, +) \triangleleft G$$

$$G/H = \{0 + H, 1 + H, \dots, 5 + H\} \cong \mathbb{Z}_6$$

Theorem

Suppose that $H \leq G$.

$$H \triangleleft G \iff \forall a \in G \ aHa^{-1} = H \iff \boxed{\forall a \in G, h \in H \ aha^{-1} \in H}$$

Theorem

Suppose that $H \leq G$.

$$H \triangleleft G \iff \forall a \in G \ aHa^{-1} = H \iff \boxed{\forall a \in G, h \in H \ aha^{-1} \in H}$$

$$\boxed{\forall a \in G \ (aH = Ha \rightarrow \forall h \in H \ aha^{-1} \in H)}$$

$$\begin{aligned} aH = Ha &\implies aHa^{-1} = (Ha)a^{-1} = H(aa^{-1}) = H \\ &\implies aHa^{-1} \subseteq H \\ &\implies \forall h \in H \ aha^{-1} \in H \end{aligned}$$

Theorem

Suppose that $H \leq G$.

$$H \triangleleft G \iff \forall a \in G \ aHa^{-1} = H \iff \boxed{\forall a \in G, h \in H \ aha^{-1} \in H}$$

$$\boxed{\forall a \in G \ (aH = Ha \rightarrow \forall h \in H \ aha^{-1} \in H)}$$

$$\begin{aligned} aH = Ha &\implies aHa^{-1} = (Ha)a^{-1} = H(aa^{-1}) = H \\ &\implies aHa^{-1} \subseteq H \\ &\implies \forall h \in H \ aha^{-1} \in H \end{aligned}$$

$$\boxed{\forall a \in G, h \in H \ (aha^{-1} \in H \rightarrow aH = Ha)}$$

Theorem

Suppose that $H \leq G$.

$$H \triangleleft G \iff \forall a \in G \ aHa^{-1} = H \iff \boxed{\forall a \in G, h \in H \ aha^{-1} \in H}$$

$$\boxed{\forall a \in G \ (aH = Ha \rightarrow \forall h \in H \ aha^{-1} \in H)}$$

$$\begin{aligned} aH = Ha &\implies aHa^{-1} = (Ha)a^{-1} = H(aa^{-1}) = H \\ &\implies aHa^{-1} \subseteq H \\ &\implies \forall h \in H \ aha^{-1} \in H \end{aligned}$$

$$\boxed{\forall a \in G, h \in H \ (aha^{-1} \in H \rightarrow aH = Ha)}$$

$$aha^{-1} \in H \implies ah = (aha^{-1})a \in Ha \implies aH \subseteq Ha$$

Theorem

Suppose that $H \leq G$.

$$H \triangleleft G \iff \forall a \in G \ aHa^{-1} = H \iff \boxed{\forall a \in G, h \in H \ aha^{-1} \in H}$$

$$\boxed{\forall a \in G \ (aH = Ha \rightarrow \forall h \in H \ aha^{-1} \in H)}$$

$$\begin{aligned} aH = Ha &\implies aHa^{-1} = (Ha)a^{-1} = H(aa^{-1}) = H \\ &\implies aHa^{-1} \subseteq H \\ &\implies \forall h \in H \ aha^{-1} \in H \end{aligned}$$

$$\boxed{\forall a \in G, h \in H \ (aha^{-1} \in H \rightarrow aH = Ha)}$$

$$\begin{aligned} aha^{-1} \in H &\implies ah = (aha^{-1})a \in Ha \implies aH \subseteq Ha \\ a^{-1}ha = a^{-1}h(a^{-1})^{-1} \in H &\implies ha \in aH \implies Ha \subseteq aH \end{aligned}$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$\forall \sigma \in S_3, \tau \in H \ \sigma \tau \sigma^{-1} \in H$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$\forall \sigma \in S_3, \tau \in H \quad \sigma\tau\sigma^{-1} \in H$$

Theorem

$$\sigma\tau\sigma^{-1} = \begin{pmatrix} \sigma(1) & \sigma(2) & \dots & \sigma(n) \\ \sigma(\tau(1)) & \sigma(\tau(2)) & \dots & \sigma(\tau(n)) \end{pmatrix}$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$\forall \sigma \in S_3, \tau \in H \ \sigma\tau\sigma^{-1} \in H$$

Theorem

$$\sigma\tau\sigma^{-1} = \begin{pmatrix} \sigma(1) & \sigma(2) & \dots & \sigma(n) \\ \sigma(\tau(1)) & \sigma(\tau(2)) & \dots & \sigma(\tau(n)) \end{pmatrix}$$

$$(1\ 2)(1\ 2\ 3)(1\ 2)^{-1} = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 3 & 2 \end{pmatrix} = (1\ 3\ 2)$$

Definition (Homomorphism (同态))

Let $(G, \cdot)$ and $(G', *)$ be two groups. Let ϕ be a function such that

$$\forall a, b \in G \quad \phi(a \cdot b) = \phi(a) * \phi(b).$$

Then ϕ is a **homomorphism** from G to G' .

Definition (Homomorphism (同态))

Let $(G, \cdot)$ and $(G', *)$ be two groups. Let ϕ be a function such that

$$\forall a, b \in G \quad \phi(a \cdot b) = \phi(a) * \phi(b).$$

Then ϕ is a **homomorphism** from G to G' .

If ϕ is a bijection, then G and G' are called **isomorphic**.

$$\phi : G \cong G'$$

$$\begin{aligned}\phi : (\mathbb{Z}, +) &\rightarrow (\mathbb{R}^*, *) \\ n &\mapsto (-1)^n\end{aligned}$$

$$\phi : (\mathbb{Z}, +) \rightarrow (\mathbb{R}^*, *)$$

$$n \mapsto (-1)^n$$

$$\phi(m+n) = (-1)^{m+n} = \phi(m)\phi(n)$$

$$\begin{aligned}\phi : (\mathbb{Z}, +) &\rightarrow (\mathbb{Z}_6, +_6) \\ a &\mapsto [a]_6\end{aligned}$$

$$\phi(a + b) = [a + b]_6 = [a]_6 +_6 [b]_6 = \phi(a) +_6 \phi(b)$$

$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

$$\phi(f(x) + g(x)) = (f(x) + g(x))' = f'(x) + g'(x) = \phi(f(x)) + \phi(g(x))$$

Theorem

Suppose that ϕ is a homomorphism from G to G' .

Let e and e' be identities of G and G' , respectively.

$$(1) \quad \phi(e) = e'$$

$$(2) \quad \phi(a^{-1}) = (\phi(a))^{-1}$$

Theorem

Suppose that ϕ is a homomorphism from G to G' .

Let e and e' be identities of G and G' , respectively.

$$(1) \quad \phi(e) = e'$$

$$(2) \quad \phi(a^{-1}) = (\phi(a))^{-1}$$

$$e'\phi(e) = \phi(e) = \phi(ee) = \phi(e)\phi(e) \implies \phi(e) = e'$$

Theorem

Suppose that ϕ is a homomorphism from G to G' .

Let e and e' be identities of G and G' , respectively.

$$(1) \quad \phi(e) = e'$$

$$(2) \quad \phi(a^{-1}) = (\phi(a))^{-1}$$

$$e'\phi(e) = \phi(e) = \phi(ee) = \phi(e)\phi(e) \implies \phi(e) = e'$$

$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = \phi(e) = e' = \phi(a)(\phi(a))^{-1}$$

Theorem

Suppose that ϕ is a homomorphism from G to G' .

(1)

$$H \leq G \implies \phi(H) \leq \phi(G)$$

(2)

$$H \triangleleft G \implies \phi(H) \triangleleft \phi(G)$$

Theorem

Suppose that ϕ is a homomorphism from G to G' .

(1)

$$H \leq G \implies \phi(H) \leq \phi(G)$$

(2)

$$H \triangleleft G \implies \phi(H) \triangleleft \phi(G)$$

(3)

$$K \leq G' \implies \phi^{-1}(K) \leq G$$

(4)

$$K \triangleleft G' \implies \phi^{-1}(K) \triangleleft G$$

$$H \triangleleft G \implies \phi(H) \triangleleft \phi(G)$$

$$H \triangleleft G \implies \phi(H) \triangleleft \phi(G)$$

$$\forall a \in G, h \in H \quad aha^{-1} \in H$$

$$H \triangleleft G \implies \phi(H) \triangleleft \phi(G)$$

$$\forall a \in G, h \in H \quad aha^{-1} \in H$$

$$\phi(a) \in \phi(G), \quad \phi(h) \in \phi(H)$$

$$H \triangleleft G \implies \phi(H) \triangleleft \phi(G)$$

$$\forall a \in G, h \in H \quad aha^{-1} \in H$$

$$\phi(a) \in \phi(G), \quad \phi(h) \in \phi(H)$$

$$\phi(a)\phi(h)\phi(a)^{-1} = \phi(a)\phi(h)\phi(a^{-1}) = \phi(aha^{-1}) \in \phi(H)$$

$$K \triangleleft G' \implies \phi^{-1}(K) \triangleleft G$$

$$K \triangleleft G' \implies \phi^{-1}(K) \triangleleft G$$

Definition (同态核 (Kernel))

Suppose that ϕ is a homomorphism from G to G' .

Let e' be the identity of G' . Then $e' \triangleleft G'$.

$$\phi^{-1}(\{e'\}) = \{a \in G \mid \phi(a) = e'\} \triangleleft G$$

is the **kernel** of ϕ , denoted $\text{Ker } \phi$.

$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\text{Ker } \phi =$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\text{Ker } \phi = 2\mathbb{Z}$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\text{Ker } \phi =$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\text{Ker } \phi = 6\mathbb{Z}$$

$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

$$\text{Ker } \phi =$$

$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

$$\text{Ker } \phi = \mathbb{R}$$

Theorem (Fundamental Homomorphism Theorem (群同态基本定理))

Suppose that ϕ is a homomorphism from G to G' . Then

$$G/\text{Ker } \phi \cong \phi(G).$$

Theorem (Fundamental Homomorphism Theorem (群同态基本定理))

Suppose that ϕ is a homomorphism from G to G' . Then

$$G/\text{Ker } \phi \cong \phi(G).$$

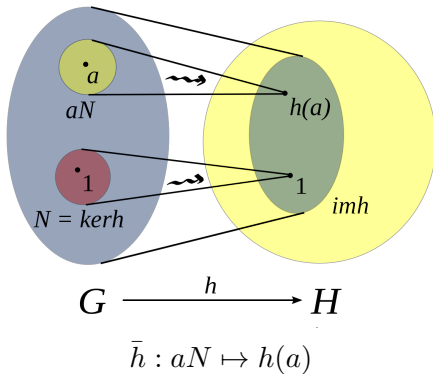
同态核可以看作群 G 与其同态像 $\phi(G)$ 之间**相似程度**的一种度量

Theorem (Fundamental Homomorphism Theorem (群同态基本定理))

Suppose that ϕ is a homomorphism from G to G' . Then

$$G/\text{Ker } \phi \cong \phi(G).$$

同态核可以看作群 G 与其同态像 $\phi(G)$ 之间**相似程度**的一种度量



$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\text{Ker } \phi = 2\mathbb{Z}$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\text{Ker } \phi = 2\mathbb{Z}$$

$$\mathbb{Z}/(2\mathbb{Z}) = (2\mathbb{Z}, 2\mathbb{Z} + 1) \cong \phi(\mathbb{Z}) = \{-1, 1\} \cong \mathbb{Z}_2$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\text{Ker } \phi = 6\mathbb{Z}$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\text{Ker } \phi = 6\mathbb{Z}$$

$$\mathbb{Z}/(6\mathbb{Z}) = \{0 + H, 1 + H, \dots, 5 + H\} \cong \phi(\mathbb{Z}) = \mathbb{Z}_6$$

$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

$$\text{Ker } \phi = \mathbb{R}$$

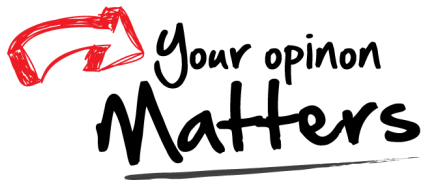
$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

$$\text{Ker } \phi = \mathbb{R}$$

$$\mathbb{R}[x]/\mathbb{R} \cong \phi(\mathbb{R}[x]) = \mathbb{R}[x]$$

Thank
You!



hfwei@hnu.edu.cn